

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic processes, likelihood-based inference, Monte Carlo simulation, numerical optimization, and scalable statistical modeling. Develops Python/PyTorch research prototypes and diagnostics for large, noisy dependent data, with finance-sector experience and demonstrated interest in market making and systematic trading.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, reproducible research pipelines, algorithm development

Quantitative Methods: Probability, stochastic processes, MLE/GLS, Monte Carlo simulation, numerical optimization, Gaussian processes, time-series and spatiotemporal modeling, spectral analysis, hypothesis and sequential testing, predictive modeling, model validation

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed and validated a Sequential Probability Ratio Test framework for binary risk indicators, modeling the cumulative log-likelihood-ratio process as a random walk with statistically controlled decision boundaries.
- Applied Brownian-motion approximation and dynamic-programming probability propagation to compute boundary-crossing probabilities, stopping-time distributions, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model validation, and predictive-performance evaluation on AWS EC2.
- Analyzed model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE RESEARCH AND TRADING

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Diagnostics

Sep 2024 - Present

- Statistical Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- Simulation and Diagnostics:** Compared Matérn and generalized Cauchy models and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and dependence.
- Research Engineering:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, Monte Carlo simulation, CPU/GPU execution, cached artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Probability Theory, Stochastic Processes, Machine Learning, Linear Algebra, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)